

Successive h-indices using OpenAlex

1 Introduction

In 2007, András Schubert published a paper titled “Successive h-indices.” (2007) It begins:

Hirsch’s excessively discussed h-index generated a long row of interpretations, applications and derivatives within a surprisingly short period. Symptomatically, Hirsch’s paper in the *Proceedings of the National Academy of Sciences of the United States of America* became the second highest cited paper of the issue in which it had been published. In this note, attention is called to an apparently so far overlooked option: h-indices themselves may form the basis of a successive series of h-indices.

Given, e.g., the h-indices of the researchers of an institute (which indices will be denoted by h_1 in what follows), an index, h_2 , of the institute can be determined: the institute has an index h_2 if h_2 of its N researchers have an h_1 -index of at least h_2 each, and the other $N - h_2$ researchers have h_1 -indices lower than h_2 each. The succession can then be continued, e.g., for networks of institutions or countries or other higher levels of aggregation.

Since no sufficient data for testing the concept on the researcher-institute-country hierarchy were available, another example was chosen.

Fifteen years later, OpenAlex (Priem, Piwowar, & Orr, 2022) launched a FOSS bibliographic catalogue complete with citation indexing, authors, fields, and institutions. Recent evaluations have found it to be a broadly suitable substitute for proprietary sources such as Scopus and Web of Science at the institutional level (Alperin, Portenoy, Demes, Larivière, & Haustein, 2024; Culbert et al., 2025), and we take it here as the dataset for which Schubert wished in 2007.

In the years between, successive h-indices attracted modest but sustained attention. Egghe (2008) gave the concept a theoretical footing, showing how the exponent of the underlying Lotka distribution propagates through each successive aggregation step. Two case studies followed shortly after: Ruane and Tol (2008) computed a rational (non-integer) successive h-index across economics departments in Ireland, and Arencibia-Jorge et al. (2008) applied Schubert’s exact researcher–department–institution hierarchy to a Cuban research institute using Web of Science data. More recently, Mryglod et al. (2022) cautioned that departmental- and institutional-level h-indices are driven largely by group size rather than quality, a critique we return to

in Section 2.6. None of this prior work, however, has had access to a comprehensive, open, institution-labeled snapshot of the kind OpenAlex now provides.

Schubert suggested that h_2 of institutes would be more interesting than that of journals; Section 2 describes how we compute it at scale, and Section 3 presents the results.

2 Methodology

2.1 Data source and filtering

The underlying data is OpenAlex’s public authors snapshot (Priem et al., 2022), distributed as Parquet on `s3://openalex/data/parquet/authors/` (us-east-1, no credentials required) and partitioned into over 1,600 daily `updated_date=` directories totalling about 53 GB. Only four columns are read from each remote file (`id`, `h_index`, `affiliations`, `topics`), and an author is retained only if all of the following hold:

- `summary_stats.h_index` is non-null and greater than zero;
- at least one entry in `affiliations` has `institution.type = 'education'`;
- at least one entry in `topics` carries a non-null field classification.

Applying these filters to the full snapshot yields 27.1 million authors, spanning 20,907 institutions and OpenAlex’s current list of 26 top-level fields. Because `s3://openalex/data/parquet/authors/` is a live, continuously-updated snapshot rather than a versioned release, these counts (and every downstream number in this paper) reflect its state as of our most recent pull; re-running the pipeline against a later snapshot state will not reproduce them exactly; we return to the size of this effect in Section 4. One limitation carried over from the source data: an author’s `h_index` is computed by OpenAlex from all works meeting its citation criteria regardless of retraction status, which OpenAlex records inconsistently (Hauschke & Nazarovets, 2025); a small, unquantified fraction of h_1 values in our dataset may therefore be inflated by since-retracted work.

2.2 Author-level assignment

Each retained author is reduced to a single row: one institution and one field. An author may list several affiliations with `institution.type = 'education'`; rather than fractionally splitting or duplicating that author across institutions, we assign them to the one with which they were most recently affiliated. Recency is read from OpenAlex’s `affiliations` field, which records, for each institution an author has ever listed, the calendar years in which they published while there; we take the institution whose most recent such year is latest, breaking ties (identical latest year) by the numerically smallest OpenAlex institution identifier. We deliberately do not use the similarly-named `last_known_institutions` field for this: despite its name, it is simply the set of affiliations listed on an author’s single most recent work, carrying no per-entry recency of its own, so when an author has more than one such affiliation (e.g., a joint appointment or co-affiliated coauthors misattributed to one author) there is no way to tell which is “more recent” than another.

Field assignment works the same way but is weighted. OpenAlex tags each author with a set of topics, each carrying a `count` (roughly, the number of the author’s works classified under that topic) and a parent field. We sum `count` within each field and assign the author to whichever field carries the largest total weight: their modal field by publication volume, not simply their single most granular topic. The result is an `authors` table with one row per author: (`author_id`, `h_index`, `institution_id`, `institution_name`, `field`, `field_name`).

2.3 The h-index and its successive generalization

Hirsch’s (2005) original index, here h_1 , is computed by OpenAlex per author from citation counts and taken as given. Schubert’s generalization (2007) lifts the same definition up a level of aggregation: a group of N members, each already carrying an h_{k-1} index, has an index $h_k = n$ if n of its members have $h_{k-1} \geq n$ and the remaining $N - n$ members have $h_{k-1} < n$:

Algorithm 1 Successive h-index of a group G , given a value $v(m)$ for each member $m \in G$

- 1: Sort G by v descending; let $r(m)$ be m ’s 1-indexed rank in that order
 - 2: **return** $\max\{r(m) : v(m) \geq r(m), m \in G\}$
-

We use this rank-based procedure, implemented as a window function in SQL ranking rows within each group and taking the largest rank whose value has not yet fallen below it, for every successive index in this paper: h_2 of an institution over the h_1 of its authors, and h_3 of a country over the h_2 of its institutions.

2.4 h_2 : institution and institution $\times$ field

h_2 is computed twice, over different groupings of the same author-level data. The field-specific table, `h2_by_institution_field`, groups authors by (institution, field) and yields 328,500 pairs; institutions are matched on their numeric identifier only; institution and field *names* are resolved afterward via `arg_max`, so that spelling or formatting variants of the same name across snapshot partitions cannot silently split one institution into several groups. The institution-overall table, `h2_by_institution`, ignores field entirely and applies the same procedure across all of an institution’s authors regardless of discipline, yielding one h_2 per institution (20,907 rows).

2.5 h_3 : country level

Institutions are mapped to countries using the `country_code` field of OpenAlex’s institutions snapshot (also read directly from S3, joined on institution ID). h_3 is then computed with the same rank-based procedure, once globally per country over the h_2 values of its institutions (`h3_by_country`), and once per (country, field) over the field-specific h_2 values (`h3_by_field/`), producing one file per field.

2.6 Supplementary metrics

Several secondary statistics are derived from the h_2 and h_3 tables to probe what the headline numbers obscure:

- **Gini coefficient**, computed on the within-field distribution of institutional h_2 , quantifies whether a field’s excellence is concentrated in one institution (Gini near 1) or broadly distributed (Gini near 0).
- **Efficiency** normalizes for scale, directly addressing the group-size dependence flagged by Mryglod et al. (2022): $h_2/\text{author_count}^{1/\beta_1}$ at the institution level, and $h_3/\text{institution_count}^{1/\beta_2}$ at the country level. Both exponents come directly from Egghe (2008): his eq. 11 gives $h_2 = \text{author_count}^{1/(\alpha_0\alpha_1)}$ and his eq. 18 gives $h_3 = \text{institution_count}^{1/(\alpha_0\alpha_1\alpha_2)}$, where $\alpha_0, \alpha_1, \alpha_2$ are the Lotka exponents of the citation-per-article, article-per-author, and author-per-institution processes respectively, so we write $\beta_1 = \alpha_0\alpha_1$ and $\beta_2 = \alpha_0\alpha_1\alpha_2$ for the two compound exponents the efficiency metrics actually need. Egghe’s own theorems (his eqs. 7 and 14) show that β_1 and β_2 are themselves the Lotka tail exponents of the h_1 -across-authors and h_2 -across-institutions distributions, so rather than assume a value for either, we fit both directly by discrete maximum likelihood (Clauset et al. (2009), via the `powerlaw` package (Alstott, Bullmore, & Plenz, 2014), with the tail threshold $x_{\min}$ chosen automatically by minimizing the Kolmogorov–Smirnov distance to the fitted distribution), giving $\beta_1 = 1.833$ (fit on all 27.1 million authors’ h_1 values) and $\beta_2 = 2.917$ (fit on all 20,907 institutions’ h_2 values). α_2 , the Lotka exponent of authors-per-institution, is fit the same way directly from the `author_count` column ($\alpha_2 = 2.021$), and α_1 , the exponent of articles-per-author, is fit from each author’s OpenAlex `works_count` ($\alpha_1 = 2.996$) after excluding 46 of 27,083,177 authors with `works_count` > 10,000: every one of these pairs an implausible publication count (up to 278,862) with a single-digit h_1 , the signature of an OpenAlex author-disambiguation failure (a generic name or collaboration byline merged into one author ID) rather than genuine output. α_0 , the citation-per-article exponent, is fit directly from a sample of 718,299 works with nonzero citations drawn from 25 OpenAlex works-snapshot partitions spread across its full 2016–2026 date range (rather than the complete ~ 725 GB snapshot, since a single ~ 360 K-row partition already yields an order of magnitude more tail data than a stable MLE fit requires), giving $\alpha_0 = 2.742$.

Taken at face value, these four independently-fit exponents are inconsistent with Egghe’s own model: $\beta_1/\alpha_1 = 0.612$ and $\beta_2/(\alpha_1\alpha_2) = 0.482$ both imply $\alpha_0 < 1$, which his model rules out by construction (eq. 4 requires every Lotka exponent to exceed 1 for the underlying population to be finite), and the predicted $\beta_2 = \beta_1\alpha_2$ comes out 21% off the observed β_2 . Section 3.11 traces this to global pooling across all 26 OpenAlex fields: refitting the same four exponents within a single field brings the implied α_0 to 0.981 for Physics and Astronomy, essentially at the boundary the model requires, while the analogous β_2 vs. $\beta_1\alpha_2$ mismatch does not shrink at all within a single field, evidence of a genuine break specifically at the institution-aggregation step rather than an artifact of heterogeneity. We report both exponents

and both findings rather than silently picking whichever normalization makes the efficiency metric look best justified.

- **Breadth score** counts, per institution, how many of the 26 fields it ranks in the global top 10, top 50, and top 100.
- **Non-biomedical ranking** recomputes institution-level h_2 after excluding authors whose modal field is Medicine, Biochemistry/Genetics/Molecular Biology, Immunology and Microbiology, Neuroscience, or Health Professions, to test whether the overall ranking is, in substance, a ranking of medical school size.
- **Academic Olympics** converts each field’s country-level h_3 ranking into an Olympic-style medal table: the top three distinct h_3 values in a field are gold/silver/bronze, ties at the same h_3 share a medal and the next tier is skipped (as in the real Olympics), and medals are tallied across all 26 fields per country.

2.7 Manual cross-validation

Because errors in OpenAlex’s underlying author disambiguation or citation counts would propagate directly into every successive index, we manually spot-check a targeted, non-random sample of authors rather than a random one, chosen at the points where an error would be both most likely and most consequential: authors sitting exactly at an institution’s h_2 threshold, where a ± 1 error in a single h_1 could change the institution’s rank; top authors in fields where OpenAlex’s topic classification is known to be thinner (e.g., Arts and Humanities); top authors at institutions with non-Anglophone naming conventions, where author-name disambiguation is more error-prone and where OpenAlex’s coverage is known to diverge unevenly from other databases by country, most severely for China (Zheng, Miao, Bu, & Larivière, 2025), including institutions suspected of affiliation-swapping for grant purposes (Bhattacharjee, 2011); and the small number of authors with the highest h_1 globally, as a sanity check at the extreme of the distribution. For each candidate we resolve the author’s OpenAlex profile through the public API and compare it by hand against Web of Science, Scopus, and Google Scholar.

3 Results

3.1 Manual cross-validation

This subsection’s comparisons were made by hand against an earlier pull of the OpenAlex snapshot and have not been redone against the current one described in Section 2; given the scale of drift documented in Section 4, the specific h_1 values quoted below should be read as illustrating the kind and rough magnitude of disagreement across sources, not as current figures for these particular authors. Of the 19 authors sampled under the scheme described in Subsection 2.7, Table 1 summarizes agreement between OpenAlex’s `h_index` and the corresponding value we located by hand in Web of Science, Scopus, and Google Scholar. Coverage itself varies sharply: Scopus returned a value for 16 of the 19 authors, Google Scholar for 12, and Web of Science for only 11.

Scopus tracks OpenAlex most closely, both in coverage and in magnitude of disagreement. Google Scholar is systematically higher than OpenAlex, consistent with

Table 1 Agreement between OpenAlex and each external source, over authors for which that source returned a value. Diff is OpenAlex minus the external value.

Source	n	Mean diff	Mean abs. diff	Worst case
Web of Science	11/19	+27.2	29.0	Lei Jiang (+100)
Scopus	16/19	+6.7	13.9	Donald Small (+46)
Google Scholar	12/19	-21.9	34.6	A. Jafari (-85)

its broader crawl of preprints, theses, and other grey literature and its comparative permissiveness toward self-citation; this ordering (Google Scholar > Scopus > OpenAlex > Web of Science) reproduces cleanly among the five global top- h_1 sanity-check authors, none of whom show a discrepancy exceeding 27% of their OpenAlex value in either direction.

Four authors, however, show disagreement well outside this range. We read this as a signature of author-disambiguation failure (multiple distinct people merged under one OpenAlex author ID) rather than of genuine cross-database disagreement:

- **Donald Small** (Johns Hopkins): $h_1^{\text{OpenAlex}} = 103$ against Scopus = 57 and Google Scholar = 59.
- **Karen S. Calhoun** (University of Michigan): $h_1^{\text{OpenAlex}} = 45$ against Scopus = 11 and Google Scholar = 13.
- **Lei Jiang** (University of Science and Technology of China): $h_1^{\text{OpenAlex}} = 215$ against Web of Science = 115, consistent with documented China-specific coverage discrepancies in OpenAlex (Zheng et al., 2025).
- **A. Jafari** (Sharif University of Technology): $h_1^{\text{OpenAlex}} = 139$, but Web of Science, Scopus, and Google Scholar report 41, 133, and 224 respectively.

All four fall in the non-Anglophone-institution or common-name strata of the sample, matching the concern raised in Subsection 2.7 that disambiguation is more error-prone for exactly these authors. Because such merges can only inflate an author’s apparent h_1 , and h_1 feeds directly into h_2 and h_3 , a handful of merged identities at the right institution could measurably distort an institution- or country-level ranking; we flag this as an open limitation rather than one this sample can size definitively.

The two Arts and Humanities threshold authors sampled (Gillian Ramchand and G. Herbert Fowler, both sitting at Oxford’s field-specific $h_2 = 30$ threshold) returned no value in any of the three external sources. Citation indexes are well documented to under-represent book-based humanities scholarship relative to journal-based fields (Nederhof, 2006); these two null results are consistent with OpenAlex inheriting the same gap, though our sample is too small to confirm it directly. Either way, h -index comparisons in these fields should be read with correspondingly less confidence than in the sciences.

3.2 The overall h_2 ranking

Table 2 lists the ten institutions with the highest institution-level h_2 , computed over all 20,907 institutions and 27.1 million authors described in Section 2.

Table 2 Top 10 institutions by overall h_2 .

Institution	h_2	Authors
Harvard University	71	92,274
University of California, Los Angeles	64	57,912
Stanford University	63	55,761
University of California, San Francisco	63	43,867
University of Washington	63	62,319
Cornell University	62	62,003
Johns Hopkins University	62	69,675
Columbia University	60	44,419
University of North Carolina at Chapel Hill	60	57,425
University of Toronto	59	75,854

Harvard’s 1st place still matches the US News and World Report ranking, but the rest of the ordering has moved further apart from it than in an earlier pull of this same pipeline: Stanford (3rd in US News) still places highly here, but Cambridge, 5th in US News, is now 26th ($h_2 = 55$), outside our own top 10 entirely. As we show below, however, the overall ranking is a considerably less interesting object than it first appears.

3.3 Field-level heterogeneity

Splitting h_2 by (institution, field) reveals that the overall ranking is deceptive: several institutions that never come close to the top of Table 2 are, in fact, the single strongest research group on Earth in their field. Table 3 lists a sample of these cases.

University of Washington is the counterexample that proves the rule holds both ways: it is #5 overall ($h_2 = 63$) and #1 in Environmental Science ($h_2 = 35$), a broad powerhouse that also happens to lead a specific, non-biomedical field outright. Two entries in Table 3 are noticeably thinner than the earlier pull of this dataset showed: USTC’s Energy lead has narrowed to a single institution ahead of Jiangsu University at $h_2 = 21$, and what was previously described as a four-field Chinese cluster (Materials Science, Engineering, Energy, Chemical Engineering) has, in the current snapshot, resolved to Tohoku University leading Materials Science, MIT leading Engineering, and Universidad de Zaragoza leading Chemical Engineering; only Energy remains USTC’s outright. Field leadership at these margins is evidently sensitive to exactly the kind of snapshot-to-snapshot drift discussed in Section 4, and should be read as a current snapshot of a moving target rather than a stable characterization of any one institution’s specialty.

Not every field supports this kind of comparison equally well. Decision Sciences is the thinnest field in the dataset: the University of Southern California’s global

Table 3 Selected field leaders that rank well outside the overall top 100. “Overall” columns give the institution’s rank and h_2 in Table 2’s full ranking; “Field h_2 ” is the institution’s h_2 restricted to that field.

Field	Leader	Ov. rank	Ov. h_2	Field h_2
Agricultural & Biological Sciences	Wageningen	137	46	40
Energy	USTC	270	41	22
Dentistry	Universidade de São Paulo	231	42	27
Chemistry	University of Bologna	153	45	33
Materials Science	Tohoku University	103	47	33
Computer Science	Carnegie Mellon University	80	48	41

#1 is $h_2 = 17$, built on 219 USC authors against 131,012 authors field-wide, versus Medicine’s $h_2 = 63$ built on 6.80 million authors field-wide. h_2 is not meaningfully comparable across fields of such different size, only within a field, and Decision Sciences in particular may be too sparse an OpenAlex topic category for its ranking to carry much signal.

3.4 Concentration within fields

Table 4 reports, for every field, the Gini coefficient of institutional h_2 defined in Section 2.6, together with the field’s institution count, leading (maximum) h_2 , and mean h_2 .

Biochemistry, Genetics and Molecular Biology is now the most unequal field, edging out Neuroscience and Physics and Astronomy: Harvard’s field-leading $h_2 = 50$ is 9.5 times the field mean of 5.29. Decision Sciences and Veterinary are, in either order, the most evenly distributed (Gini ≈ 0.39 , essentially tied). Medicine sits in the middle of the table (0.484) despite Harvard’s large absolute lead there too ($h_2 = 63$ against a field mean of 6.77), because Medicine, unlike the most concentrated fields above it, has thousands of institutions with meaningful h_2 rather than a single dominant outlier.

3.5 The overall ranking as an implicit biomedical ranking

Medicine alone accounts for 6.80 million of the dataset’s 27.1 million authors (25%), and, together with Biochemistry/Genetics/Molecular Biology, Immunology and Microbiology, Neuroscience, and Health Professions, the five core biomedical fields dwarf most other fields’ author pools. Table 5 recomputes institution-level h_2 after excluding authors in these five fields entirely to test whether the overall ranking in Table 2 is, in substance, a ranking of medical school size.

Unlike in an earlier pull of this pipeline, Harvard does *not* remain #1 once its biomedical authors are removed: it falls to 5th, overtaken by Cornell, Stanford, Berkeley, and UCLA. That is, if anything, a cleaner version of this section’s argument than a

Table 4 Gini coefficient of institutional h_2 within each field, sorted from most to least concentrated.

Field	Institutions	Max h_2	Mean h_2	Gini
Biochemistry, Genetics & Molecular Biology	15,109	50	5.29	0.534
Neuroscience	11,381	42	3.65	0.529
Physics and Astronomy	12,203	40	4.39	0.526
Immunology and Microbiology	9,610	41	3.30	0.517
Materials Science	12,801	33	5.15	0.517
Chemistry	11,902	33	4.38	0.512
Earth and Planetary Sciences	12,125	37	3.50	0.504
Engineering	17,633	42	5.76	0.494
Environmental Science	16,382	35	4.42	0.488
Medicine	19,042	63	6.77	0.484
Agricultural and Biological Sciences	16,252	40	4.65	0.482
Computer Science	16,649	41	4.44	0.480
Psychology	15,628	32	3.12	0.468
Dentistry	8,151	27	2.68	0.465
Mathematics	10,915	26	3.05	0.460
Energy	8,762	22	2.76	0.451
Economics, Econometrics and Finance	13,782	28	2.91	0.445
Business, Management and Accounting	14,522	22	3.16	0.443
Social Sciences	19,183	32	4.06	0.432
Health Professions	14,774	27	2.58	0.429
Pharmacology, Toxicology and Pharmaceutics	7,052	19	2.33	0.421
Arts and Humanities	15,284	24	2.56	0.420
Chemical Engineering	5,748	16	2.21	0.407
Nursing	8,872	19	2.10	0.390
Veterinary	4,506	22	1.88	0.386
Decision Sciences	10,232	17	2.14	0.385

surviving #1 would have been: Harvard’s overall #1 position (Table 2) depends partly on the scale of its medical school, not purely on cross-field excellence, since even Harvard drops four places once that scale is set aside. Berkeley, MIT, and UIUC show the pattern most starkly, gaining 18, 27, and 37 places respectively once Medicine and its four sibling fields are excluded, while UCLA, Washington, and Columbia, all anchored by large medical schools of their own, fall 2 to 3 places. Caltech and ETH Zurich show the same pattern one tier down (16th and 17th non-biomedical against 49th and 50th overall, $\Delta\text{rank} = +33$ each), and University of Science and Technology of China further down still (112th non-biomedical against 270th overall, $\Delta\text{rank} = +158$). Because medical fields have by far the largest, most-cited author pools, they structurally dominate any university-wide aggregate; an institution elite in non-medical fields, such as USTC or Carnegie Mellon, can never reach the top of the overall list no matter how good it is, simply because it lacks a large medical school. The field-level and non-biomedical breakdowns matter more than the unrestricted university-level number.

Table 5 Top 10 institutions by non-biomedical h_2 , alongside their position in the unrestricted overall ranking. Δ rank is overall rank minus non-biomedical rank; positive means the institution moves up once biomedical fields are excluded.

Institution	Rank	h_2	Overall rank	Overall h_2	Δ rank
Cornell University	1	53	6	62	+5
Stanford University	2	53	3	63	+1
University of California, Berkeley	3	52	21	56	+18
University of California, Los Angeles	4	52	2	64	-2
Harvard University	5	51	1	71	-4
Massachusetts Institute of Technology	6	51	33	53	+27
University of Illinois Urbana-Champaign	7	51	44	52	+37
University of Washington	8	51	5	63	-3
University of Wisconsin-Madison	9	51	18	57	+9
Columbia University	10	50	8	60	-2

3.6 Breadth versus depth

Table 3 showed institutions that lead a single field while sitting far outside the overall top 100. Table 6 shows the opposite pattern: institutions whose strength is broad rather than deep, measured by how many of the 26 fields they place in the global top 50.

Table 6 The eight institutions with the most top-50 field placements out of 26 fields.

Institution	Top 50	Top 10	Overall rank	Overall h_2
University of Minnesota	20	4	17	57
University of Wisconsin-Madison	20	4	19	57
Harvard University	19	12	1	71
University of Toronto	19	8	10	59
Cornell University	18	10	6	62
University of California, Los Angeles	18	10	2	64
University of Washington	18	10	5	63
University of Florida	17	3	45	52

Minnesota and Wisconsin-Madison now tie for the greatest breadth by this measure, placing in the global top 50 in 20 of 26 fields despite leading only 4 outright each, more consistently broad than even Harvard’s 19. Harvard still leads more fields outright (5 of 26) than any other institution, but Minnesota and Wisconsin place fewer than a quarter of their 20 top-50 fields inside the top 10, meaning essentially all of their breadth comes from being solidly good across many fields rather than dominant in any one; Toronto shows the same pattern less extremely (8 of 19). At the other extreme, several institutions post a high overall h_2 without ever leading a single field in the top 10: University of Michigan-Ann Arbor (rank 16, $h_2 = 57$), Washington University in St. Louis (rank 24, $h_2 = 56$), and the University of Arizona (rank 32,

$h_2 = 54$) all have zero top-10 field placements. These institutions achieve their overall strength through scale and consistency across many merely-good departments, not through fielding the single best group in any one discipline.

3.7 Institution efficiency and the β_1 scaling law

Section 2.6 derived $h_2 = \text{author_count}^{1/\beta_1}$ from Egghe’s eq. 11 with $\beta_1 = 1.833$ fit by discrete MLE. Figure 1 plots this relationship directly: a regression of h_2 on $\text{author_count}^{1/\beta_1}$, forced through the origin, across all 18,569 institutions with at least 10 authors gives slope 0.229 and $R^2 = 0.809$ — a somewhat weaker fit than the $\sqrt{N}$ heuristic it replaces ($R^2 = 0.88$), visibly so at the largest institutions in Figure 1, where the fitted line runs above the bulk of the data. $\beta_1 = 1.833$ is, after all, an empirical tail exponent of the h_1 distribution pooled across every field and career stage in OpenAlex, not a free parameter tuned to make this particular regression look good; Section 3.11 shows that same pooling is exactly what makes β_1 an imperfect description of any single, more homogeneous sub-population.

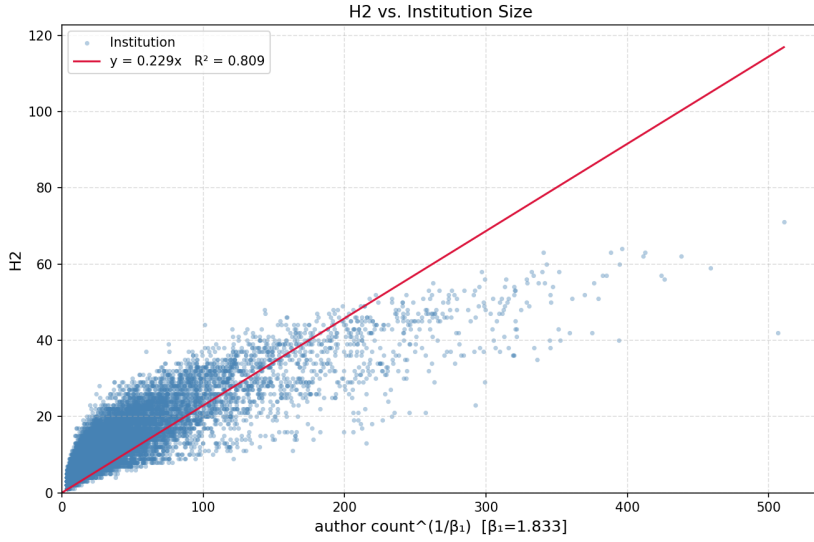


Fig. 1 Institution-level h_2 versus $\text{author_count}^{1/\beta_1}$ ($\beta_1 = 1.833$), with a least-squares fit forced through the origin ($R^2 = 0.809$, $n = 18,569$).

Because the fit is in expectation, institutions that beat it are unusually efficient: they achieve a given h_2 with far fewer authors than the trend predicts. Table 7 lists the five most efficient institutions by $h_2/\text{author_count}^{1/\beta_1}$ among institutions with at least 100 authors. All five have well under 600 authors, and the overlap between the two rankings is empty: not one of the institutions in the overall top 100 by raw h_2 also appears in the top 100 by efficiency. Depth of talent and efficient use of a small author pool are, in this dataset, close to mutually exclusive achievements. Two of the five, however, illustrate the same caution as Saudi Arabia’s h_3 efficiency in Subsection 3.10:

Parkway School District (a K-12 district, not a research institution) and Kyung Hee Cyber University (a distance-learning institution) reaching a double-digit h_2 on barely over 100 authors each looks more like a small, idiosyncratic author pool than genuine depth of talent, a risk this metric runs at low author counts regardless of exponent.

Table 7 Top 5 institutions by $h_2/\text{author_count}^{1/\beta_1}$ (minimum 100 authors).

Institution	Efficiency	h_2	Authors	h_2 rank
Parkway School District	1.27	19	142	2,940
Kyung Hee Cyber University	1.22	17	126	3,637
Center for Coastal Studies	1.21	15	101	4,447
University Centre in Svalbard	1.18	16	119	4,289
Royal Society of Edinburgh	1.17	15	107	4,732

3.8 h_3 : country level

Aggregating institutional h_2 up to the country level via the same rank-based procedure yields h_3 . Table 8 lists the ten highest- h_3 countries, and Figure 2 maps h_3 for all 214 countries with at least one qualifying institution.

Table 8 Top 10 countries by h_3 (11 rows shown: Australia and South Korea tie for 9th).

Country	h_3	Institutions
United States	49	4,239
Germany	37	505
United Kingdom	37	547
China	35	1,078
Italy	34	228
Japan	34	1,356
Spain	31	156
France	30	394
Canada	29	304
Australia	28	120
South Korea	28	358

The United States leads by a wide margin, but Italy’s 34 on only 228 institutions and Japan’s 34 on 1,356 are a first hint that raw h_3 rewards national research-system size as well as depth, a point to which we return in Subsection 3.10. Germany and the United Kingdom tie for 2nd at 37 despite the UK having 42 more institutions; Australia and South Korea tie for 9th at 28 despite South Korea having exactly three times as many institutions (358 vs. 120), a starker version of the same pattern.

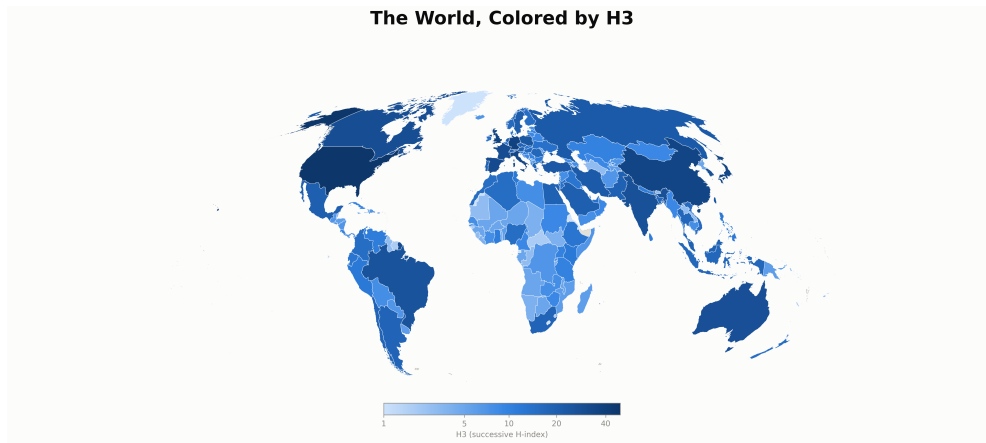


Fig. 2 h_3 by country. Countries with no qualifying institution are shown in grey.

3.9 Academic Olympics

Section 2.6 defined an Olympic-style medal table by awarding gold/silver/bronze to the top three distinct h_3 tiers in each of the 26 fields, with ties sharing a medal. Figure 3 shows the resulting medal table.

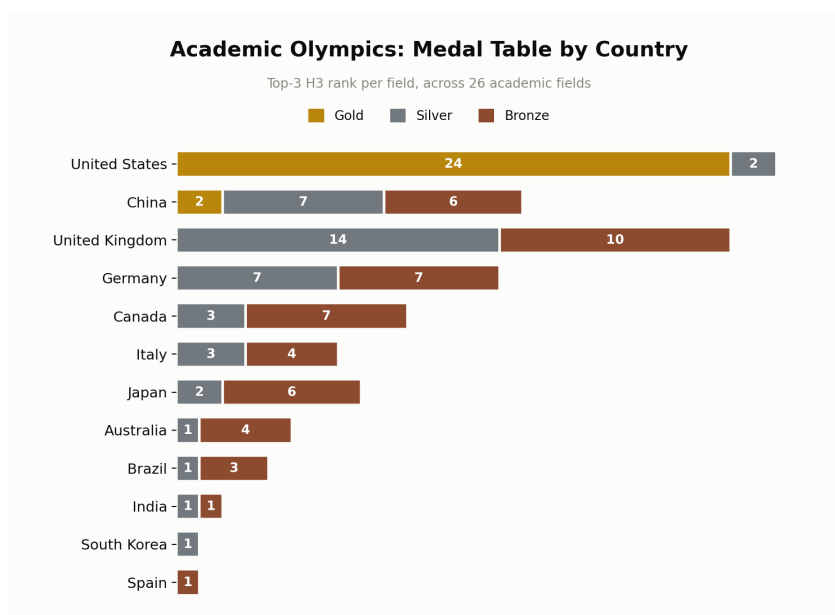


Fig. 3 Academic Olympics medal table: gold/silver/bronze by top-3 h_3 rank per field, across 26 fields.

The United States now wins 24 of 26 golds, with China holding the remaining 2 (Energy, Materials Science). This is a substantially different picture from an earlier pull of this same pipeline, which had the US at 22 golds and China at 5 (Chemical Engineering, Chemistry tied, Energy, Engineering, Materials Science); Section 4 discusses this drift at the dataset level, but three of these five races are close enough to be worth reporting exactly rather than as a clean reversal: Chemical Engineering flipped to the US by a single institution (US $h_3 = 10$ vs. China 9), Materials Science remains China’s by an equally thin margin (China 25 vs. US 24), and only Engineering (US 30 vs. China 28) and Chemistry (US 23 vs. China 20, no longer a tie) moved by more than one point. Energy remains China’s most comfortable applied-science gold (China 16 vs. the runner-up South Korea’s 12). Whatever this snapshot’s medal count, the underlying finding survives: China’s strength is concentrated in applied and materials science, with zero medals in arts, humanities, social sciences, or business. Great Britain is the clearest case of depth without a peak: 14 silvers and 10 bronzes, but zero golds, consistent with world-class institutions that are nonetheless outgunned by US depth in every field.

3.10 Country efficiency and the β_2 scaling law

Section 2.6 derived $h_3 = \text{institution_count}^{1/\beta_2}$ from Egghe’s eq. 18 with $\beta_2 = 2.917$ fit by discrete MLE. Figure 4 plots this relationship directly: a regression of h_3 on $\text{institution_count}^{1/\beta_2}$, forced through the origin, across the 151 countries with at least 5 institutions gives slope 3.137 and $R^2 = 0.937$ — a markedly better fit than the $\log_2 N$ heuristic it replaces ($R^2 = 0.84$).

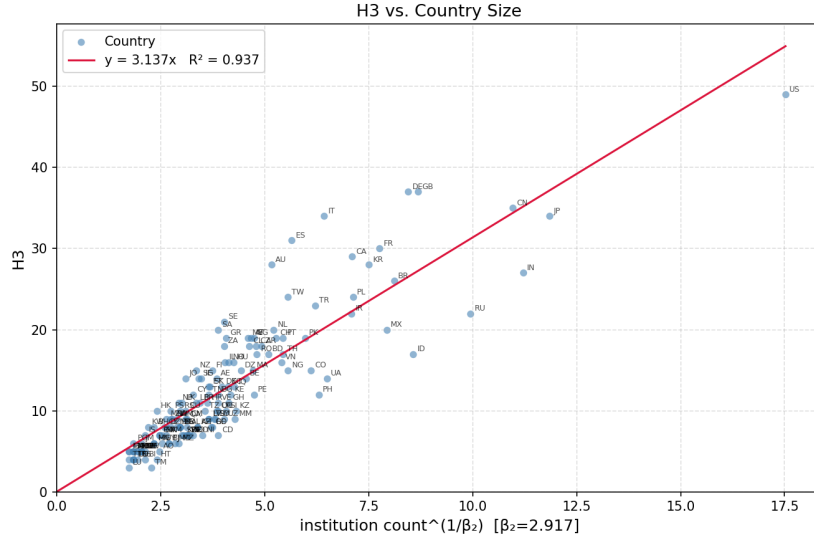


Fig. 4 Country-level h_3 versus $\text{institution_count}^{1/\beta_2}$ ($\beta_2 = 2.917$), with a least-squares fit forced through the origin ($R^2 = 0.937$, $n = 151$).

Table 9 ranks countries by $h_3/\text{institution_count}^{1/\beta_2}$. The change of normalization matters here far more than it did for h_2 : because $\text{institution_count}^{1/\beta_2}$ eventually grows faster than $\log_2(\text{institution_count})$, it penalizes large research systems much more heavily, and the United States, #1 by the old log-based efficiency measure, falls all the way to #84 (efficiency 2.80) despite still leading every country on raw h_3 . Spain now leads at 5.49, followed closely by Australia (5.42) and Italy (5.29); the next four (Sweden, Saudi Arabia, Greece, Jordan) all reach h_3 in the 14–21 range on a few dozen institutions. Saudi Arabia’s continued presence near the top (5th, $h_3=20$ on 52 institutions) is worth the same caution as before: as flagged in Subsection 2.7, this is plausibly an artifact of high- h -index researchers listing Saudi institutions as secondary affiliations in exchange for funding (Bhattacharjee, 2011), rather than a genuine concentration of resident research talent.

Table 9 Top 10 countries by $h_3/\text{institution_count}^{1/\beta_2}$ (minimum 5 institutions).

Country	Efficiency	h_3	Institutions	h_3 rank
Spain	5.49	31	156	7
Australia	5.42	28	120	10
Italy	5.29	34	228	5
Sweden	5.22	21	58	19
Saudi Arabia	5.16	20	52	22
Greece	4.67	19	60	26
Jordan	4.52	14	27	51
New Zealand	4.48	15	34	47
South Africa	4.47	18	58	33
Germany	4.38	37	505	2

3.11 Testing the compounding-exponent model

Section 2.6 noted that the four exponents fit globally are inconsistent with Egghe’s own model: β_1/α_1 and $\beta_2/(\alpha_1\alpha_2)$ both imply $\alpha_0 < 1$, which his eq. 4 rules out. Egghe’s derivation assumes a single, homogeneous Lotka process at each level; fitting globally pools together every OpenAlex field’s citation practices at once, from Medicine to Arts and Humanities, which is a very different population from the one his model describes. To test whether this pooling is what breaks the model, we refit all four exponents restricted to a single field at a time. Table 10 shows the result for Medicine (the largest field, 6.80 million authors) and Physics and Astronomy (a mid-sized field, 756,898 authors, culturally a very different citation regime from Medicine).

The two rows tell different stories. Implied α_0 moves steadily toward the model’s required boundary as the population narrows: 0.612 globally, 0.890 within Medicine, 0.981 within Physics and Astronomy, essentially consistent with $\alpha_0 = 1$ once field heterogeneity is controlled for. That supports treating the author-level violation as an artifact of pooling citation practices from every field into one fit, not a flaw in the compounding logic itself. The β_2 row does not resolve the same way: the mismatch

Table 10 Implied $\alpha_0 = \beta_1/\alpha_1$ and the β_2 vs. $\beta_1\alpha_2$ compounding check, global vs. within a single field. A model-consistent fit requires implied $\alpha_0 > 1$ and $\beta_2 \approx \beta_1\alpha_2$.

	Global (all fields)	Medicine	Physics and Astronomy
Implied $\alpha_0 = \beta_1/\alpha_1$	0.612	0.890	0.981
β_1 vs. power law	avored	exponential favored	exponential favored
β_2 vs. $\beta_1\alpha_2$ predicted	−21%	−42%	−45%
β_2 vs. power law	exponential favored	exponential favored	exponential favored

between observed β_2 and the model’s predicted $\beta_1\alpha_2$ does not shrink within a single field, and in both fields tested the h_2 -across-institutions distribution is more consistent with an exponential tail than a power-law one – a result that recurs identically at the global level. Unlike the author-level story, this looks like a genuine break in Egghe’s model specifically at the institution-aggregation step (his eq. 14), not an artifact of heterogeneity. We use the empirically fit β_1 and β_2 in the efficiency metrics regardless: Egghe’s eq. 11 and eq. 18 functional forms are a defensible normalization independent of whether this deeper multiplicative-consistency check holds, and Figures 1 and 4 show both fits are still reasonable descriptions of h_2 and h_3 as functions of group size, whatever the shape of the underlying tail.

4 Conclusion

Schubert closed his 2007 note wondering whether h_2 of institutes would prove more interesting than h_2 of journals, absent the researcher-institute-country data needed to check. Nineteen years and 27.1 million OpenAlex authors later, the answer is yes, but not in the way a single top-10 table would suggest: the overall ranking in Table 2 turns out to be substantially a ranking of biomedical author-pool size (Subsection 3.5, where even Harvard’s own #1 position does not survive excluding biomedical fields), and it is only once h_2 is split by field (Subsection 3.3) that the successive index proves valuable, surfacing institutions such as Wageningen, Tohoku, and the University of Bologna as world leaders in specific disciplines invisible to any university-wide metric. Egghe’s (2008) compounding-exponent model gives the efficiency metrics of Subsections 3.7 and 3.10 real theoretical footing in place of the $\sqrt{N}/\log_2 N$ heuristics of an earlier version of this analysis: both $h_2 = \text{author_count}^{1/\beta_1}$ and $h_3 = \text{institution_count}^{1/\beta_2}$ use exponents fit directly from the data by discrete MLE rather than assumed, and both hold reasonably well ($R^2 = 0.809$ and 0.937 respectively) across several orders of magnitude in group size. Subsection 3.11’s direct test of Egghe’s model went further than the efficiency metrics strictly required and found a genuinely mixed result: the model’s author-level consistency requirement, violated when fit globally, is restored almost exactly once field heterogeneity is controlled for, while its institution-level requirement is not, a specific and reproducible boundary on where a 2008 Lotkaian model of citation aggregation still holds against 2026 data.

None of this is airtight, and one limitation is not of the usual kind. `s3://openalex/data/parquet/authors/` is a live, continuously-updated snapshot, not a versioned release, and partway through this analysis we had to fully re-pull it to

add one column (`works_count`, needed to fit α_1). That re-pull did not reproduce the authors snapshot an earlier version of this paper was computed from: overall author count fell from roughly 30 million to 27.1 million, Harvard’s institution-level h_2 fell from 137 to 71 on a similarly reduced author count (92,274 vs. 110,979 previously), and, more strikingly, several field-level claims reversed outright rather than merely shifting – University of Antwerp no longer leads Physics and Astronomy, the University of Chinese Academy of Sciences no longer leads Chemical Engineering, and China’s Academic Olympics gold count fell from 5 to 2 (Subsection 3.9), though three of those five field-level races are decided by margins of one or two institutions and could plausibly flip again on the next pull. We cannot fully disentangle how much of this is OpenAlex correcting author-disambiguation errors of exactly the kind Subsection 3.1 independently flags (huge, implausible `works_count` values paired with single-digit h -indices, which we excluded via a fixed cutoff before fitting α_1 in Subsection 2.6) versus ordinary snapshot growth versus some other backend change, since OpenAlex does not version its public snapshot in a way that lets us diff the two states directly. Whatever the cause, every number in this paper is a snapshot of a moving target, and a live, continuously-updated source is a genuine hazard to point-in-time bibliometric claims at this level of aggregation, not just a hypothetical one: we could reproduce this fact but not the specific numbers it produced eighteen months from now with the identical pipeline.

The cross-validation in Subsection 3.1 (against an earlier pull, per the note there) found four of nineteen sampled authors with h_1 discrepancies large enough to suggest author-disambiguation error rather than genuine cross-database disagreement and concentrated exactly where expected: non-Anglophone institutions and common names. Because such errors propagate upward by construction, a handful of merged identities at the right institution could shift an h_2 or h_3 ranking, and we have no way to size that effect from a sample chosen for likely error rather than representativeness. Coverage gaps likely compound this in fields like Arts and Humanities, where citation databases are well documented to under-represent book-based scholarship (Nederhof, 2006) and our own cross-validation sample turned up no external value at all for either Arts and Humanities author checked. A natural next step would be to reassign h_2 to apply to departments rather than universities, once OpenAlex or another database provides such data. Then, h_3 could be calculated at the university level, and h_4 at the country level. The results here still seem, in Schubert’s own terms, conceivable but not bluntly trivial: a positive sign for the successive h -index as a coherent frame for multi-level research assessment, nineteen years and several orders of magnitude of data on from where he left it – provided the specific numbers are read as a snapshot, not a fixed measurement.

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